

```
File Edit Sketch Tools Help
sketch_nov09a $
const char* ssid = "beral"; //Define your own SSID
const char* password = " "; // Define your own password
int pulse=14; //GPIO14 to act as trigger pin
int head=1; //Define the threshold number of object capture
#include <WiFi.h>
#include <WiFiClientSecure.h>
#include "esp_camera.h"
#include "soc/soc.h"
#include "soc/rtc_cntl_reg.h"

String Feedback="";
String Command="",cmd="",P1="",P2="",P3="",P4="",P5="",P6="",P7="",P8="",P9=""; // bounday parameters of object

byte ReceiveState=0,cmdState=1,strState=1,questionstate=0,equalstate=0,semicolonstate=0;
```

Fig. 4: Setting Wi-Fi SSID and password in code

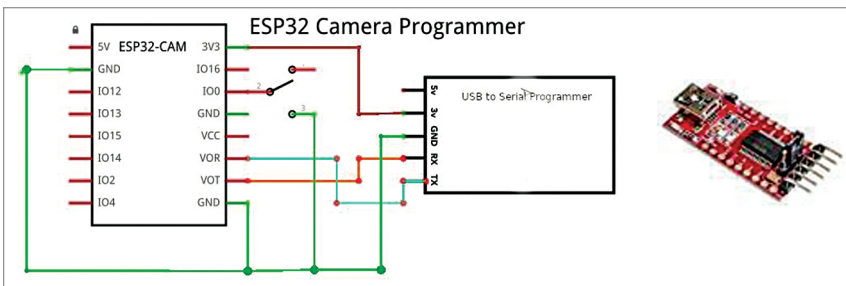


Fig. 5: ESP32 cam programming circuit

10 seconds) that is built around IC 555.

Once triggered, it keeps the relay on for a minimum of 1 to 10 seconds, depending on the preset value of the 100k variable resistor. The trigger pin of IC 555 is its pin number 2, which is connected to the collector of transistor BC547 vide GPIO14. If the GPIO14 is high, it connects the trigger pin 2 to the ground through emitter of the transistor.

Pin details of ESP32 cam are shown in Fig. 3.

The main part of the project is object detection and an alert system. So, the relay circuit can be replaced with an optocoupler relay module and the regulator circuit can be replaced with a 5V adaptor like a mobile phone charger.

Software

For preparing the code for this project please ensure the latest Arduino IDE 1.8 is installed. Install the Arduino IDE on ESP32 board using instructions given on the esp website and then set the SSID and password in the code. Source code setting screenshot is

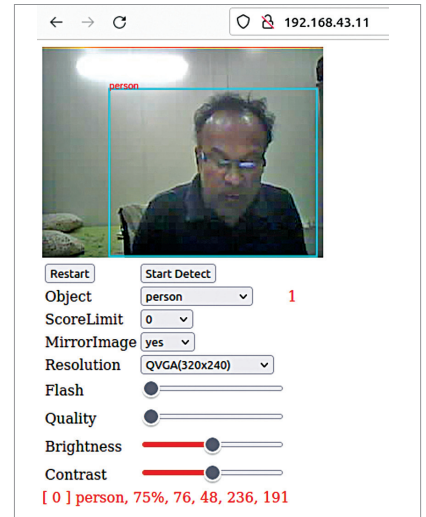


Fig. 6: ESP32 cam output detecting the person and showing results

address of the device in the browser.

The coco-SSD model used in the project can detect a wide range of objects, which you can select from the pull-down menu that has various objects—from a person to a tooth-brush (see Fig. 6).

Whatever object is selected, if the number is more than 1, or as decided by you, the ESP32 cam will monitor that number. If it exceeds, it will command the relay to switch on the alarm circuit.

The following additional project ideas can be tried for this project; your imagination is the limit:

- **Scarecrow.** When birds, dogs, cats, cows, cars, etc come near it will sound an alarm.

- **Automatic magic eye.** Can be installed at the front door so that when somebody comes it rings the bell automatically, but is not activated by stray dogs, cats, etc.

- **Guests counter.** Can be installed where people gather—such as in a temple, mosque, or banquet hall—to count the number of people entering from the door where it is installed. **EFY**

shown in Fig. 4.

EFY note. You can download the whole code from the link <https://www.electronicsforu.com/electronics-projects/object-detection-esp32-camera-acts-as-a-sentry>

The programming circuit of ESP cam is shown in Fig. 5. The cam does not come with built-in USB-to-serial interface shown on the right side of Fig. 5. Therefore, to upload the sketch, a USB-to-serial programmer is required. The GPIO 0 is to be pulled to ground during uploading and the reset button of ESP32 should be pressed during the first few seconds of uploading; after that it is to be released.

Construction and testing

Upload the source code in ESP32 cam and assemble the circuit. Power the device with 5V and search the IP

EFY Note

The source code of this project is available for free download at www.electronicsforu.com



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